Absolutely maximally entangled states of 15 and 17 parties with local dimension eight, and the automorphism group of a stabilizer AME(8,4) state

Claude Opus 5.5 Extra Effort[1]

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An n-party pure state of local dimension d is absolutely maximally entangled, AME(n, d), if every set of $\lfloor n/2 \rfloor$ parties is maximally mixed. Which pairs (n, d) admit such states is open in general. We construct AME$(15, 8)$ and AME$(17, 8)$, two cases that were open in the table of known AME states. Both are graph states whose adjacency matrix is a symmetric circulant over $\mathbb{F}_8$. They were verified by exact rank computations over $\mathbb{F}_8$ and over $\mathbb{F}_2$ with three independent programs. An exhaustive scan of all symmetric circulants over $\mathbb{F}_8$ finds 336 and 48 solutions up to scaling, in 28 and 4 orbits under multipliers and field automorphisms. For AME$(8, 4)$, the smallest open case, we show the following. A stabilizer AME$(8, 4)$ state with respect to any group-type nice error basis must be an $\mathbb{F}_2$-linear (qubit-pair) stabilizer state. Such a state, if it exists, has a trivial automorphism group in $\mathrm{GL}(4, 2) \wr S_8$. Each prime-order symmetry sector is excluded either by a counting argument or by an exhaustive search; all searches but one were implemented twice. Hence no symmetric ansatz yields a group-type stabilizer AME$(8, 4)$. The results were obtained by a multi-agent language-model system working against a frozen exact verifier.

I. INTRODUCTION

A pure state $|\psi\rangle \in (\mathbb{C}^d)^{\otimes n}$ is *absolutely maximally entangled*, AME(n, d), if $\rho_A = \mathbb{1}/d^{|A|}$ for every set A of $|A| = \lfloor n/2 \rfloor$ parties. Equivalently it is a perfect tensor, or a pure quantum code $((n, 1, \lfloor n/2 \rfloor + 1))_d$. AME states are the resources of quantum secret sharing and parallel teleportation [1], and the building blocks of holographic codes [2]. They are closely tied to orthogonal arrays and MDS codes [3, 4]. Their existence is a long-standing open problem (problem 35 of the IQOQI list; see the review [5]). Nonexistence results come from the shadow inequalities [6, 7], Scott's bound [8], and special arguments for four and seven qubits [9, 10]. Constructions come mostly from stabilizer codes and quantum MDS codes [4, 11, 12], and from a few non-stabilizer states such as the "golden" AME$(4, 6)$ [13].

The Huber–Wyderka table [14] records the status of every cell with $d \leq 11$. Three 2026 preprints settled further cells: all seven-party cases $d \geq 3$ [15], AME$(12, 5)$, AME$(18, 11)$ and AME$(18, 13)$ via Hermitian self-dual MDS codes [16], and AME$(2q, q)$ for every odd prime power q [17]. A non-stabilizer AME$(9, 6)$ was found recently, with a Lean proof in its author's repository [18, 19]. The kernel of Ref. [17] has no direct analogue for $q = 2^r \geq 8$ (their Lemma S4), and the even prime powers $d = 4, 8$ remain the least understood. The first open case overall is AME$(8, 4)$ [5, 15].

This paper reports two results. Section III constructs AME$(15, 8)$ and AME$(17, 8)$. Both cells were open (Table I); the code tables list $[[15, 0, d]]_8$ with $7 \leq d \leq 8$ and $[[17, 0, d]]_8$ with $8 \leq d \leq 9$, while AME needs $d = 8$ and $d = 9$ [20]. The states are qudit graph states with circulant symmetry over $\mathbb{F}_8$, and a simple counting heuristic suggests why these two cells, and no neighbouring ones, fall to this method. Section IV turns to AME$(8, 4)$. We reduce every group-type stabilizer formalism for ququarts to qubit pairs, and we show that a stabilizer AME$(8, 4)$ state would have to be completely asymmetric (Theorem 3). Section V describes how the results were obtained and checked.

TABLE I. Existence of AME(n, d) for $13 \leq n \leq 22$ and $d \in \{4, 5, 7, 8, 9, 11\}$, after this work ($d = 2$ and 3 are omitted; there AME$(15, 3)$ is the only open cell in this range). Sources: the Huber–Wyderka table [14] (updated 8 June 2026) and the 2026 preprints [15–17]. ✓ known to exist, ★ this work, × known not to exist, ? open. AME$(19, 11)$ and AME$(20, 11)$ are announced, without a construction, in Ref. [18]. Over all cells with $2 \leq d \leq 11$ up to Scott's bound, the snapshot has 117 existing, 96 nonexistent and 907 open cells.

$d\backslash n$	13	14	15	16	17	18	19	20	21	22
4	?	?	?	×	?	?	?	×	?	?
5	✓	✓	✓	?	✓	✓	?	?	?	?
7	✓	✓	✓	?	✓	✓	?	?	?	?
8	✓	✓	★	?	★	?	?	?	?	?
9	✓	✓	✓	✓	✓	✓	?	?	?	?
11	✓	✓	✓	✓	✓	✓	?	?	✓	✓

II. STABILIZER AME STATES AND GRAPH STATES

Let $d = q = p^m$. A qudit stabilizer state on n parties is described by an $\mathbb{F}_p$-linear subspace $S \subset \mathbb{F}_p^{2mn}$ of dimension mn. Each party contributes m coordinates for X and m for Z, and S is isotropic for the symplectic form $\omega((a|b), (a'|b')) = a \cdot b' - b \cdot a'$ [21]. For a set A of parties, $\rho_A = d^{-|A|} \sum_{g \in S,\ \mathrm{supp}\, g \subseteq A} \chi(g)\, g|_A$ for a character χ, so ρ_A is maximally mixed exactly when no nonzero element of S is supported inside A. Writing S as the row space of an $mn \times 2mn$ matrix M, the elements supported in

A form a space of dimension $mn - \operatorname{rank} M[\,\cdot\,, A^c]$, where $M[\,\cdot\,, A^c]$ keeps the X and Z columns of the parties outside A. This gives the criterion used throughout:

Lemma 1. *The stabilizer state of S is AME(n, p^m) if and only if, for every set B of $\lceil n/2 \rceil$ parties, the $mn \times 2m\lceil n/2 \rceil$ column block $M[\,\cdot\,, B]$ has rank mn over $\mathbb{F}_p$.*

An $\mathbb{F}_q$-linear graph state is given by a symmetric matrix $\Gamma \in \mathbb{F}_q^{n\times n}$ with zero diagonal. For $q = 2^m$, write Tr for the trace $\mathbb{F}_q \to \mathbb{F}_2$. The state is

$$|\Gamma\rangle = q^{-n/2} \sum_{x\in\mathbb{F}_q^n} (-1)^{\mathrm{Tr}\left(\sum_{i<j} \Gamma_{ij} x_i x_j\right)} |x\rangle, \qquad (1)$$

stabilized by $X(\lambda e_i)Z(\lambda\Gamma_i)$ for $\lambda \in \mathbb{F}_q$, where $X(a)|x\rangle = |x + a\rangle$ and $Z(b)|x\rangle = (-1)^{\mathrm{Tr}(b\cdot x)}|x\rangle$ [22–24]. An element $\sum_i X(c_i e_i)Z(c_i\Gamma_i)$ is supported in A only if $c_i = 0$ for $i \notin A$ and $c_A\Gamma[A, A^c] = 0$. Hence:

Lemma 2. *$|\Gamma\rangle$ is AME(n, q) if and only if $\Gamma[A, A^c]$ has full row rank $\lfloor n/2 \rfloor$ over $\mathbb{F}_q$ for every set A of $\lfloor n/2 \rfloor$ parties.*

Expanding $\mathbb{F}_q$ in an $\mathbb{F}_p$-basis for the X part, and in its trace-dual basis for the Z part, turns the $\mathbb{F}_q$-symplectic structure into the $\mathbb{F}_p$-symplectic one of Lemma 1 [21]. This is the form our verifier checks. Projecting one party of an AME$(2k, d)$ state onto a basis state gives an AME$(2k-1, d)$ state [5, 17]. The converse step, from odd to even n, does not hold in general.

III. AME(15,8) AND AME(17,8)

Write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$. A symmetric circulant on $\mathbb{Z}_n$ is $\Gamma_{ij} = c_{\delta(i,j)}$ with $\delta(i, j) = \min(|i-j|, n-|i-j|)$ and $c = (c_1, \ldots, c_{\lfloor n/2 \rfloor})$.

Theorem 1. *The graph states (1) with symmetric circulant Γ over $\mathbb{F}_8$ and*

$$n = 15: \quad c = (1,\ 0,\ \alpha^2,\ \alpha^4,\ \alpha^3,\ 0,\ 0),$$
$$n = 17: \quad c = (1,\ \alpha,\ \alpha,\ 1,\ \alpha,\ \alpha^6,\ \alpha^6,\ \alpha)$$

are AME$(15, 8)$ and AME$(17, 8)$, respectively. Equivalently, they are pure stabilizer codes $[[15, 0, 8]]_8$ and $[[17, 0, 9]]_8$.

Verification. By Lemma 2, it suffices to check that all 6,435 matrices $\Gamma[A, A^c]$ (7×8) for $n = 15$, and all 24,310 matrices (8×9) for $n = 17$, have full row rank over $\mathbb{F}_8$. We did this with three programs that share no code, except that (iii) reuses the verifier's $\mathbb{F}_2$ row reduction to compare row spaces. (i) The frozen verifier of the search (Appendix A) checks the $3n$-qubit $\mathbb{F}_2$ stabilizer matrix stored in the candidate file: isotropy and rank in Python, all $\binom{n}{\lceil n/2 \rceil}$ column blocks of Lemma 1 by Gaussian elimination in C, and a seeded sample of blocks again in Python. (ii) An auditing agent wrote its own $\mathbb{F}_2$ bit-vector elimination over all blocks, together with an $\mathbb{F}_8$-level rank check of every $\Gamma[A, A^c]$. As a negative control for $n = 15$, exchanging one qubit between two parties produced 268 failing sets. (iii) The script accompanying this paper rebuilds Γ from the coefficients above with its own $\mathbb{F}_8$ arithmetic, checks every $\Gamma[A, A^c]$, and confirms that its $\mathbb{F}_2$ expansion spans the same stabilizer group as the certified candidate files. □

How they were found. The search scanned *all* symmetric circulants over $\mathbb{F}_8$ with first nonzero coefficient 1: 299,593 for $n = 15$ (336 AME states) and 2,396,745 for $n = 17$ (48 AME states). Each scan takes seconds in C. Every hit was re-verified. The maps $i \mapsto ui$ ($u \in \mathbb{Z}_n^\times$, a party permutation), $c \mapsto c^2$ (a local relabelling, since $\mathrm{Tr}(z^2) = \mathrm{Tr}(z)$), and $c \mapsto \lambda c$ (local scaling, since every $\lambda \in \mathbb{F}_8^\times$ is a square) preserve the AME property. On normalized vectors, scaling is absorbed and the group generated by multipliers modulo ± 1 and Frobenius has order 12 for $n = 15$ and 24 for $n = 17$. The solutions fall into 28 orbits for $n = 15$ and 4 for $n = 17$, all of size 12. For $n = 17$ this is half the group order: every solution, including the one in Theorem 1, is invariant under the multiplier $i \mapsto 4i \pmod{17}$, so its party-permutation symmetry contains $\mathbb{Z}_{17} \rtimes \mathbb{Z}_4$. Orbits may still be locally equivalent to one another, so these are upper bounds on the number of inequivalent states in the family.

Why these two cells. For a random $\mathbb{F}_q$-linear $[[n, 0]]_q$ stabilizer code, the expected number of nonzero stabilizers of party weight at most $\lfloor n/2 \rfloor$ is $E = \sum_{w=1}^{\lfloor n/2 \rfloor} \binom{n}{w}(q^2-1)^w/q^n$. Dividing by the $q-1$ scalar multiples and the $2n$ dihedral images gives $\lambda = E/(2n(q-1))$, the expected number of "bad orbits" in a symmetric circulant. Table II lists λ for the scanned cells. Hits occur at $\lambda = 3.5$ and 11.4, and no AME state appears in the exhaustive scans with $\lambda \geq 20$. The Poisson guess $e^{-\lambda}$ is a rough guide, and its calibration depends on the normalization. With the dihedral factor $2n$ it overestimates the $n = 15$ hit rate by a factor of about 30 and predicts about 27 hits for AME$(13, 4)$, where the scan finds none. With a factor n instead, it predicts about 290 hits for $n = 15$ (336 found) and fewer than one for AME$(13, 4)$. Even n is much harder than the neighbouring odd n: λ is 51 for $(16, 8)$ against 3.5 and 11 for $(15, 8)$ and $(17, 8)$. This is why symmetric circulants cannot reach AME$(16, 8)$, although both of its neighbours now exist. Nor does an odd-n state project down to the even case (Sec. II). Known cells such as AME$(14, 7)$ and AME$(18, 7)$ are realized by Paley-type (quadratic-residue, bordered) circulants, where algebraic structure beats the first-moment count.

TABLE II. Exhaustive scans of symmetric circulant graph states over $\mathbb{F}_q$ (first coefficient normalized). $E/(q-1)$ is the expected number of low-weight stabilizers up to scalars in a random $\mathbb{F}_q$-linear code, and $\lambda = E/(2n(q-1))$. "Circulants" is the family size $(q^{\lfloor n/2 \rfloor}-1)/(q-1)$, every member of which was tested.

(n,q)	$E/(q-1)$	λ	circulants	AME found
(15,8)	104	3.5	299,593	336 hits (28 orbits)
(17,8)	388	11.4	2,396,745	48 hits (4 orbits)
(16,8)	1,644	51.4	2,396,745	0
(19,8)	1,450	38.2	19,173,961	0
(16,7)	1,853	57.9	960,800	0
(19,7)	1,859	48.9	6,725,601	0
(16,5)	2,410	75.3	97,656	0
(19,5)	3,311	87.1	488,281	0
(19,9)	1,159	30.5	48,427,561	0
(19,11)	785	20.7	235,794,769	0
(13,4)	102	3.9	1,365	0
(15,4)	360	12.0	5,461	0

IV. STABILIZER AME(8,4)

A. Which stabilizer formalisms remain

$\mathbb{F}_4$-linear stabilizer states (Pauli operators $X(a)Z(b)$, $a,b \in \mathbb{F}_4^8$) cannot be AME$(8,4)$: the code tables give $d \le 4$ for $[[8,0,d]]_4$ [20], and we re-derive this below. A ququart, however, admits other error bases. Let each party carry a group-type nice error basis $\{E_g\}_{g\in G}$, $|G| = 16$, with $E_gE_h = \omega(g,h)E_{gh}$ [25, 26]. A stabilizer state is the unique joint eigenvector of $\{E_g : g \in K\}$ for a subgroup $K \le G^8$ of order 4^8. As above, ρ_A is maximally mixed if and only if no $g \neq e$ in K is supported in A.

Proposition 1. *If such a state is AME$(8,4)$, then $G \cong \mathbb{Z}_2^4$, and K is a Lagrangian subspace S of $V = \bigoplus_{j=1}^8 \mathbb{F}_2^4$ for the sum of the local symplectic forms (a qubit-pair stabilizer state) in which every nonzero vector has party weight at least 5.*

Proof. For any set T of four parties, the projection $\pi_T : K \to \prod_{j\in T} G_j$ has as kernel the elements supported on the other four parties, so it is injective. Since $|K| = 16^4$, it is an isomorphism. Identify $K = G_1 \times \cdots \times G_4$ through $\pi_{\{1,2,3,4\}}$. The coordinate of party 5 is then a homomorphism $\prod_k \alpha_k(g_k)$, where the $\alpha_k : G_k \to G_5$ have pairwise commuting images. Each α_k is injective, by bijectivity of $\pi_{\{1,2,3,4,5\}\setminus\{k\}}$, hence surjective. So $G_5 = \alpha_1(G_1)$ commutes with $\alpha_2(G_2) = G_5$, and G_5 is abelian. By symmetry every G_j is abelian, and cancellation in $G_u \times G_T \cong G_v \times G_T$ makes them isomorphic. An abelian index group of a nice error basis is of symmetric type $A \times A$ with a nondegenerate alternating bicharacter [26]. With $|G| = 16$, this means $\mathbb{Z}_2^4$ or $\mathbb{Z}_4^2$. For $\mathbb{Z}_4^2$ (Weyl–Heisenberg over $\mathbb{Z}_4$), $K \cong \mathbb{Z}_4^8$ is free, because π_T is a bijection onto $\mathbb{Z}_4^8$. Its reduction $\bar{K} = K \bmod 2 \subset (\mathbb{F}_2^2)^8$ then has dimension 8, it is isotropic because commutation mod 4 implies commutation mod 2, and $2s \in K$ has the party support of $\bar{s}$. Thus $\bar{K}$ would be a stabilizer AME$(8,2)$ state, which does not exist [6, 8]. □

A related Singleton-type argument shows more generally that a $\mathbb{Z}_{p^2}$ stabilizer AME(n,p^2) requires a stabilizer AME(n,p). From now on, $S \subset V$ is a qubit-pair Lagrangian with minimum party weight 5, and $\pi_T(S) = \mathbb{F}_2^{16}$ for every set T of four parties. Taking $T = \{1,2,3,4\}$ gives the normal form $S = \{(u, Mu)\}$ with $M \in \mathrm{Sp}(16,2)$, in which all 69 party-aligned square blocks of M are invertible. These conditions encode the 35 balanced cuts.

B. Local symmetries

Let $\mathrm{Aut}(S)$ be the group of $\Omega = P_\sigma\, g$ ($\sigma \in S_8$ permutes parties, $g \in \mathrm{GL}(4,2)^8$) with $\Omega S = S$, and $\mathrm{Aut}_{\mathrm{loc}}(S)$ its subgroup with $\sigma = 1$. Because S projects onto every set of at most four parties, an automorphism acting trivially on S is the identity.

Theorem 2. $\mathrm{Aut}_{\mathrm{loc}}(S) = 1$.

Proof sketch. Since $|\mathrm{GL}(4,2)| = 2^6 \cdot 3^2 \cdot 5 \cdot 7$, it suffices to exclude a local $\Omega \neq 1$ of prime order ℓ. For odd ℓ, $S = S^\Omega \oplus S_{\mathrm{nf}}$, where S^Ω lies in the fixed spaces $U_j = \ker(\omega_j - 1)$ and $S_{\mathrm{nf}} \neq 0$ lies in $W_j = \ker \Phi_\ell(\omega_j)$. Both summands have minimum party weight ≥ 5, so deleting any four parties is injective on each. This Singleton-type bound, applied to all assignments of per-party types $(\dim U_j, \dim W_j)$, leaves only $(2,2)^8$ and $(0,4)^8$ for $\ell = 3$, $(0,4)^8$ for $\ell = 5$, and $(1,3)^8$ for $\ell = 7$.

- $(2,2)^8$: S_{nf} is an $\mathbb{F}_4$-linear $[8,4,5]_4$ MDS code, which does not exist [27].
- $(1,3)^8$: S^Ω is a binary code of length 8, dimension ≥ 4 and distance ≥ 5, impossible since $A(8,5) = 4$.
- $(0,4)^8$: S is $\mathbb{F}_4$-linear for a fixed-point-free $\mathbb{F}_4$-structure ω_j on every party. Up to local $\mathrm{Sp}(4,2)$ there are three types: H (ω_j symplectic, 40 elements), S (ω_j self-adjoint, 12), and M (60). Mixed H/S types split S into a product across a cut, which is impossible. Type M makes S a module over a non-commutative chain ring of order 16. A Singleton argument on its socle again forces an $[8,4,5]_4$ MDS code, and a similar count excludes M parties mixed with H or S parties. Uniform H means S is a Hermitian self-dual $[16,8]_4$ code with paired coordinates. Uniform S is the $\mathbb{F}_4$-linear case of the code tables. Both uniform cases were excluded by exhaustive depth-first searches over systematic generator matrices $[I|M]$ over $\mathbb{F}_4$ with all party-aligned minors invertible ($MM^\dagger = I$ for type H). Each

search was written twice, with different normalizations. The searches were calibrated on $n = 4$ and $n = 6$, where they reproduce independent brute-force counts (for example, 34 992 of the 77 760 unitary 4×4 matrices over $\mathbb{F}_4$), and against random sampling at intermediate depths. The best H-type state has one bad cut out of 35.

For $\ell = 2$, $\omega_j - 1$ is nilpotent with rank $r_j \in \{0, 1, 2\}$, and the same counting leaves $(1)^8$ and $(2)^8$. In the first case $(\Omega - 1)S$ is a binary $[8, \geq 4, \geq 5]$ code, impossible. In the second, $(\Omega - 1)S$ is an orthogonal array $\mathrm{OA}(256, 8, 4, 4)$ of index one, which violates Bush's bound $k \leq t + 1$ for $t \geq s$ [28]. The counting parts of this proof were written by one agent and have not been re-derived independently; its exhaustive components (uniform H and S) were implemented twice. □

C. Party-permuting symmetries

Theorem 3. *A qubit-pair stabilizer AME*$(8, 4)$ *state has* $\mathrm{Aut}(S) = 1$ *in* $\mathrm{GL}(4, 2) \wr S_8$.

Proof outline. (a) *Every automorphism is symplectic.* For $\Omega \in \mathrm{Aut}(S)$ the adjoint $\Omega^\dagger$ also preserves $S = S^\perp$. So $\Omega^\dagger \Omega = g^\dagger g$ is a local automorphism and equals 1 by Theorem 2, i.e. every $g_j \in \mathrm{Sp}(4, 2)$. (b) *Reduction to prime order.* If $\mathrm{Aut}(S) \neq 1$, it contains an element $\Omega = P_\sigma g$ of prime order $\ell \in \{2, 3, 5, 7\}$, and $\sigma \neq 1$ by Theorem 2. Since $\Omega^\ell = 1$, the product of the local maps around each cycle of σ is trivial. A local change of basis therefore makes Ω a pure permutation on the moved parties, with local maps $h_j \in \mathrm{Sp}(4, 2)$, $h_j^\ell = 1$, on the fixed parties. Because $\mathrm{Sp}(4, 2) \cong S_6$ has no element of order 7, $h = 1$ when $\ell = 7$. (c) *All sectors are empty.* Table III lists every cycle type of σ and every class of twists h_j. Each sector is excluded by a counting argument or by an exhaustive computation. All exhaustive computations except the 7-cycle enumeration were implemented twice with independent code. The 7-cycle enumeration was validated against direct enumeration of the 2^{16} stabilizer elements. Each sector has a positive control, in most cases a relaxed weight threshold of 4 that produces solutions. □

Corollary 1. *No qubit-pair stabilizer AME*$(8, 4)$ *is invariant under any nontrivial group of party permutations combined with local linear maps. In particular, cyclic, dihedral,* $\mathrm{AGL}(1, 8)$, $\mathrm{A\Gamma L}(1, 8)$, $\mathrm{PSL}(2, 7)$ *and* $\mathrm{AGL}(3, 2)$*-symmetric ansätze cannot succeed, and by Proposition 1 neither can any other group-type stabilizer formalism.*

D. What remains

For stabilizer states, only the asymmetric sector remains: a matrix $M \in \mathrm{Sp}(16, 2)$ with 69 invertible block minors and no symmetry. The set of all qubit-pair Lagrangians has at least about 2^{46} orbits under $\mathrm{Sp}(4, 2) \wr S_8$. A naive first-moment estimate gives about 2^{-17} expected orbits of solutions. The same model gives 2^{-132} for $\mathrm{AME}(10, 4)$, however, and $\mathrm{AME}(10, 4)$ exists, so the estimate only says that unstructured solutions would be a coincidence. Simulated annealing in graph form over 17 175 restarts reached a total rank deficit of 2, never 0 (a random graph has a deficit of about 30). For general non-stabilizer states, random-start continuous solvers fail even where AME states are known to exist ($\mathrm{AME}(6, 4)$, $\mathrm{AME}(4, 6)$). The ratio of their plateau to the random baseline is 0.345 at $(8, 3)$, where no AME state exists, and 0.350 at $(8, 5)$, where one does, so a plateau at $(8, 4)$ carries no information.

Other cells. Hermitian self-dual GRS codes do not exist for $n > q + 1$ [30]. $\mathrm{AME}(16, 8)$ is equivalent to a Hermitian self-orthogonal MDS $[15, 7, 9]_{64}$ code (shorten a self-dual $[16, 8, 9]_{64}$ code). Using the Ball–Vilar criterion [31] and an enumeration in a trace code, we found rigorously that no truncated GRS $[15, 7]_{64}$ code is Hermitian self-orthogonal. Exhaustive searches over many further structured families found no new states. These families are Hermitian codes from pure and bordered group circulants (cyclic, $\mathbb{Z}_3^2$, $\mathbb{Z}_2^3$, $\mathbb{Z}_2 \times \mathbb{Z}_4$, and modular group algebras in characteristic 2), constacyclic and twisted codes, elliptic codes, and graph codes with abelian Cayley, multiplier, orbital and twisted symmetry. The graph-code families covered the open cells with $13 \leq n \leq 20$ and $d \in \{3, 4, 5, 7, 8, 9, 11\}$; the Hermitian families ran on selected even-n cells up to $n = 24$. The repository lists them with sizes.

V. HOW THE RESULTS WERE OBTAINED

The search was run by a swarm of five language-model agents (Claude Sonnet 5.5) for four hours on one laptop, coordinated through a message board. Permission rules prevented the agents from modifying the verifier, the record data and the configuration, and the protocol forbade editing each other's files. Only verdicts of the frozen verifier counted as results. The verifier is pinned by hash, and its tests include known AME states, every verdict path and brute-force cross-checks. An auditing agent re-checked both new states and the computational sectors of Sec. IV with independent code, wrote out most of the proofs summarized there, and searched the literature. Theorem 2 and the 7-cycle sector are the auditor's own work and lack an independent re-derivation. Two red-team rounds (88 adversarial inputs in the second) produced no wrongly accepted candidate. The two new states were verified 3.3 and 4.1 minutes after the start, and the $\mathrm{AME}(8, 4)$ classification took the rest of the run. Scripts, notebooks, the board record and all certificates are in the accompanying repository. This manuscript was drafted by a language model (Claude Opus 5.5) from those records and checked against them.

TABLE III. The prime-order sectors of $\mathrm{Aut}(S)$ for a qubit-pair stabilizer AME$(8,4)$, $S = S^{\Omega} \oplus S_{\mathrm{nf}}$ (odd ℓ) or $(\Omega-1)S \subseteq S^{\Omega}$ ($\ell = 2$). "Twist" is the local map h_j on the parties fixed by σ. Each counting row has a hand derivation, and each computational row has two independent implementations, except the $\ell = 7$ row (one implementation).

ℓ	σ (twist)	reduction and result
any	1	Theorem 2.
2	(2^4)	$S^{\Omega} = (\Omega-1)S = \{(w,w) : w \in C\}$ with C an AME$(4,4)$ qubit-pair code. All AME$(4,4)$ codes form a single orbit of $\mathrm{Sp}(4,2)^4 \rtimes S_4$: there are 4 478 976 000 of them, equal to the group order divided by the 1440 automorphisms of a representative. S is then determined by a symmetric form β on C; all 2^{36} forms give minimum weight ≤ 4.
2	$(2^3,1^2)$	$S^{\Omega} = (\Omega-1)S$ is determined by a pair of forms and an isometry: 8640 candidates, 10 up to gauge. A lift search for the rest of S finds 0 solutions (at threshold 4 it finds 1.4×10^6).
2	$(2^2,1^4)$	S^{Ω} requires a nondegenerate subspace among the 432 subspaces of $(\mathbb{F}_2^2)^4$ with block weight ≥ 3; all 432 are degenerate.
2	$(2,1^6)$	Counting forces equal twist ranks $r \leq 2$. $r = 0$ and $r = 1$ fail by weight counting. $r = 2$ would need an OA$(16,6,4,2)$, which violates the Bose–Bush bound [29].
3	$(3,1^5)$, any	Counting over all 3^5 twist patterns: none survives.
3	$(3^2,1^2)$, $h = 1$	S_{nf} is the graph of one of the $\lvert\mathrm{GU}(4,2)\rvert = 77\,760$ $\mathbb{F}_4$-isometries; all have weight-4 words.
3	$(3^2,1^2)$, h with 2-dim fixed space	1152 completions T of S_{nf}, none admits a valid S^{Ω}.
3	$(3^2,1^2)$, mixed	Parity and weight counting.
3	$(3^2,1^2)$, h fixed-point-free	391 910 400 pairs (X_A, X_B) reduce to 7 classes and 138 completions T; none of the 720 $f \in \mathrm{Sp}(4,2)$ gives S^{Ω} (best: 3 bad cuts).
5	$(5,1^3)$, order-5 h	Counting forces two twisted fixed parties. S_{nf} is then one of the $333\,125 = (q+1)(q^3+1)(q^5+1)$ ($q = 4$) generators of the Hermitian polar space $H(5,16)$; the maximum minimum weight is 4.
7	$(7,1)$	$S_{\mathrm{nf}} = L \oplus L^{\perp}$ for an $\mathbb{F}_8$-subspace $L \subseteq \mathbb{F}_8^4$; all 5917 such L give minimum weight ≤ 4.

VI. CONCLUSION

AME$(15,8)$ and AME$(17,8)$ exist, as circulant graph states over $\mathbb{F}_8$. AME$(16,8)$ is now the gap between two existing cells. If a stabilizer AME$(8,4)$ exists, it is a qubit-pair state without any symmetry. Two natural next steps are an exhaustive classification of the asymmetric sector in the normal form $M \in \mathrm{Sp}(16,2)$, which would decide whether AME$(8,4)$ can be a stabilizer state at all, and structured non-stabilizer constructions for AME$(8,4)$ in the spirit of Refs. [13, 19].

Appendix A: The verifier

Candidates are either stabilizer generator matrices over $\mathbb{F}_p$ (m qudits per party) or explicit state vectors with amplitudes in $\mathbb{Z}[\zeta_N]/\sqrt{D}$. For stabilizers, the verifier checks isotropy and rank. It then checks Lemma 1 for all $\binom{n}{\lceil n/2 \rceil}$ sets in C, re-checks a seeded sample (all sets when small) in Python, and confirms for even n that every set and its complement agree (all sets when $\binom{n}{n/2} \leq 2 \times 10^5$, otherwise a sample). For states, the identities $d^a \sum_c a_{(r,c)} \overline{a_{(s,c)}} = D\,\delta_{rs}$ are evaluated at every primitive N-th root of unity modulo primes $P \equiv 1 \pmod N$. Since $\mathbb{Z}[\zeta_N]/(P) \cong \mathbb{F}_P^{\varphi(N)}$, vanishing at all roots means every coefficient is divisible by P. Taking primes whose product exceeds twice a rigorous coefficient bound then makes the identity exact. Verdicts compare the cell with a frozen snapshot of the AME table and the 2026 preprints.

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